consists of a fixed coil and moving iron piece. The fixed coil is wounded on a former. The fixed coil carries the current proportional to the voltage to be measured or the current to be measured. The moving iron is a flat disc which is eccentrically mounted on the spindle. The spindle is supported between the jewel bearings. A pointer is also attached to the spindle which moves over a graduated scale.

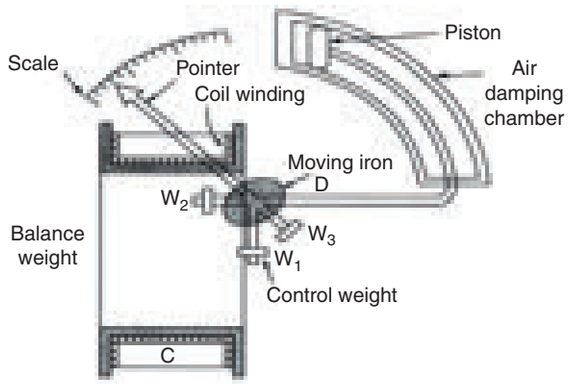


Fig. 6.7 Attraction type moving iron instrument

The basic principle of these instruments is that when a soft iron piece is brought near the magnet gets attracted by the magnet. When the current flows through the coil, a magnetic field is produced. Thus the moving iron gets attracted and moves into the coil, causing the spindle and the pointer to rotate.

The controlling torque is provided by the springs but gravity control may be used for vertically mounted panel type instruments. Damping torque is provided by air friction. A light aluminium piston is attached to the moving system. It moves in a fixed air chamber which is closed at one end. The movement of piston provides the required damping torque.

(ii) Repulsion type moving iron instruments

Repulsion type moving iron instrument is shown in figure 6.8. It has two vanes inside the coil, one fixed and the other movable. The vanes are radial stripes of iron. The fixed vane is attached to the coil and the movable vane to the spindle of the instrument.

The fixed coil carries the current or the current proportional to the voltage to be measured. When the current flows through the coil, the two vanes are magnetized in the same direction and there is a force of repulsion between the two vanes resulting in the motion of moving vane. As the moving iron is fixed with the spindle, the pointer moves on the scale when the spindle moves. The controlling torque is provided by springs and the damping torque is produced by Air friction. Whatever may be the direction of current in the coil, the two vanes are magnetized in the same direction. So the spindle always moves in one direction. Therefore, these instruments can be used on both AC and DC.

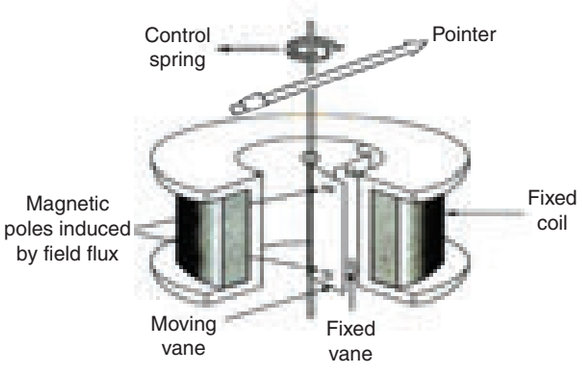


Fig. 6.8 Repulsion type moving iron instrument

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